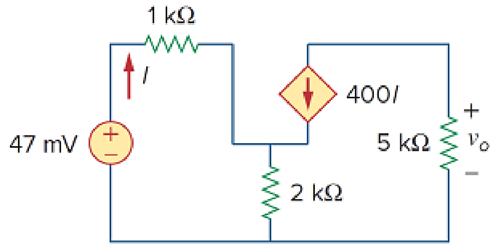
Chapter 3, Problem 86P

Problem

For the simplified transistor circuit of Fig. 3.122, calculate the voltage *vo*.

Figure 3.122:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.122 in the circuit.

Draw the circuit diagram with loop currents.

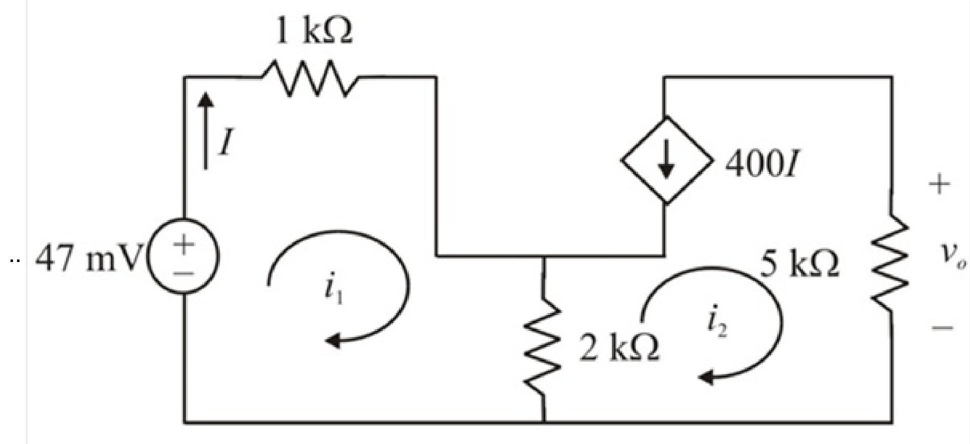


Figure 1

Step 2 of 3

From Figure 1, it is clear that,

$$I = i_1$$

$$400I = -i_2$$
$$400i_1 = -i_2$$
$$i_2 = -400i_1$$

Apply Kirchhoff's voltage law to loop1.

$$-\left(47\times10^{-3}\right)+\left(1\times10^{3}\right)i_1+\left(2\times10^{3}\right)\left(i_1-i_2\right)=0$$
$$3\times10^{3}i_1-2\times10^{3}i_2=47\times10^{-3}$$
$$3000i_1-2000i_2=47\times10^{-3}$$
$$3i_1-2i_2=47\times10^{-6}$$

Substitute $-400i_1$ for i_2.

$$3i_1-2\left(-400i_1\right)=47\times10^{-6}$$
$$803i_1=47\times10^{-6}$$

$$\begin{aligned} i_1 &= \frac{47\times10^{-6}}{803} \\ &= 58.53\times10^{-9} \\ &= 58.53\text{ nA} \end{aligned}$$

Step 3 of 3

Calculate the value of the current, i_2.

$$i_2 = -400i_1$$

Substitute 58.53 nA for i_1.

$$\begin{aligned} i_2 &= -400\left(58.53\times10^{-9}\right) \\ &= -23.412\times10^{-6} \\ &= -23.412\ \mu\text{A} \end{aligned}$$

Calculate the output voltage.

$$v_o = i_2\left(5000\right)$$

Substitute $-23.412\ \mu\text{A}$ for i_2.

$$\begin{aligned} v_o &= \left(-23.412\times10^{-6}\right)\left(5000\right) \\ &= -117.06\times10^{-3} \\ &= -117.06\text{ mV} \end{aligned}$$

Thus, the output voltage is $\boxed{-117.06\text{ mV}}$.